Astro (Trees)

Astro is an insecticide used for tree and shrub applications and must be diluted with water according to label directions before use. Mix thoroughly and maintain a uniform suspension during application. Apply using a low-pressure sprayer or similar equipment directly to the foliage, branches, and trunk of target trees, ensuring thorough coverage without runoff. Avoid spraying when rain is expected within 24 hours, and do not apply to edible crops unless the label allows. Reapply according to label intervals for continued pest control. Astro works on contact and systemic activity, providing effective control of chewing and sucking insects on trees and shrubs.

SAFETY DATA SHEET
ASTROÒ INSECTICIDE

SDS # : 1784-1-A
Revision date: 2019-06-03
Format: NA
Version 1.04

1. PRODUCT AND COMPANY IDENTIFICATION

Product Identifier

Product Name ASTROÒ INSECTICIDE

Other means of identification

Product Code(s) 1784-1-A

Synonyms PERMETHRIN (FMC 33297): 3-phenoxybenzyl (1RS,3RS;1RS,3SR)-3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropanecarboxylate or 3-phenoxybenzyl (1RS)-cis-trans-3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropanecarboxylate (IUPAC name); (3-phenoxyphenyl)methyl 3-(2,2-dichloroethenyl)-2,2-dimethylcyclopropanecarboxylate (CAS name)

Active Ingredient(s) Permethrin

Chemical Family Pyrethroid Pesticide

Alternate Commercial Name Dragnet FT, Dragnet 36.8 EC Termiticide/Insecticide

Recommended use of the chemical and restrictions on use

Recommended Use: Insecticide

Restrictions on Use: Use as recommended by the label.

Supplier Address

FMC Corporation
2929 Walnut Street
Philadelphia, PA 19104
(215) 299-6000 (General Information)
msdsinfo@fmc.com (E-Mail General Information)

Emergency telephone number

Medical Emergencies :
1 800 / 331-3148 (U.S.A. & Canada)
1 651 / 632-6793 (All Other Countries - Collect)

For leak, fire, spill or accident emergencies, call:
1 800 / 424-9300 (CHEMTREC - U.S.A.)
1 703 / 741-5970 (CHEMTREC - International)
1 703 / 527-3887 (CHEMTREC - Alternate)

2. HAZARDS IDENTIFICATION

Classification

OSHA Regulatory Status

This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200)

Acute toxicity - Oral	Category 4
Acute toxicity - Inhalation (Dusts/Mists)	Category 4

Skin sensitization	Category 1B
Specific target organ toxicity (single exposure)	Category 2
Specific target organ toxicity (repeated exposure)	Category 2
Aspiration toxicity	Category 1

GHS Label elements, including precautionary statements

EMERGENCY OVERVIEW

Danger

Hazard Statements

H302 - Harmful if swallowed
H304 - May be fatal if swallowed and enters airways
H317 - May cause an allergic skin reaction
H332 - Harmful if inhaled
H371 - May cause damage to organs
H373 - May cause damage to organs through prolonged or repeated exposure



Precautionary Statements - Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling
P270 - Do not eat, drink or smoke when using this product
P272 - Contaminated work clothing should not be allowed out of the workplace
P280 - Wear protective gloves
P260 - Do not breathe dust/fume/gas/mist/vapors/spray

Precautionary Statements - Response

P308 + P311 - If exposed or concerned: Call a POISON CENTER or doctor
P302 + P352 - IF ON SKIN: Wash with plenty of water and soap
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention
P363 - Wash contaminated clothing before reuse
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P312 - Call a POISON CENTER or doctor if you feel unwell
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor
P331 - Do NOT induce vomiting
P330 - Rinse mouth

Precautionary Statements - Storage

P405 - Store locked up

Precautionary Statements - Disposal

P501 - Dispose of contents/container to an approved waste disposal plant

Hazards not otherwise classified (HNOC)

No hazards not otherwise classified were identified.

Other Information

Very toxic to aquatic life with long lasting effects. Causes mild skin irritation. May be harmful in contact with skin.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Family Pyrethroid Pesticide.

Chemical name	***CAS-No***	***Weight %***
Permethrin	52645-53-1	36.8
Petroleum distillates, hydrotreated light	64742-47-8	10-20
1,1'-Biphenyl, bis(1-methylethyl)-	69009-90-1	20-30

Synonyms are provided in Section 1.

4. FIRST AID MEASURES

Eye Contact Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Call a poison control center or doctor for further treatment advice.

Skin Contact Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for further treatment advice.

Inhalation Move to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Ingestion Immediately call a poison control center or doctor. Do not induce vomiting unless told to do so by a poison control center or doctor. Do not give any liquid to the person. Do not give anything by mouth to an unconscious person.

Most important symptoms and effects, both acute and delayed Central nervous system effects.

Indication of immediate medical attention and special treatment needed, if necessary This product is a pyrethroid. If large amounts have been ingested, the stomach and intestines should be evacuated. Treatment is symptomatic and supportive. Digestible fats, oils, or alcohol may increase absorption and so should be avoided. Contains petroleum distillate. Vomiting may cause aspiration pneumonia.

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media Foam. Carbon dioxide (CO2). Dry chemical. Water spray or fog.

Specific Hazards Arising from the Chemical Combustible liquid

Explosion data
Sensitivity to Mechanical Impact No information available.
Sensitivity to Static Discharge No information available.

Protective equipment and precautions for firefighters As in any fire, wear self-contained breathing apparatus and full protective gear. Isolate fire area. Evaluate upwind.

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions Isolate and post spill area. Remove all sources of ignition. Wear suitable protective clothing, gloves and eye/face protection. For personal protection see section 8.

Other For further clean-up instructions, call FMC Emergency Hotline number listed in Section 1 "Product and Company Identification" above.

Environmental Precautions Keep people and animals away from and upwind of spill/leak. Keep material out of lakes, streams, ponds, and sewer drains. Keep out of waterways.

Methods for Containment Dike to prevent runoff. Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal.

Methods for cleaning up

Clean and neutralize spill area, tools and equipment by washing with water and soap.

Absorb rinsate and add to the collected waste. Waste must be classified and labeled prior to recycling or disposal. Dispose of waste as indicated in Section 13.

7. HANDLING AND STORAGE

Handling

Do not contaminate other pesticides, fertilizers, water, food, or feed by storage or disposal.

Storage

Keep in a dry, cool and well-ventilated place. Keep away from open flames, hot surfaces and sources of ignition. Keep out of reach of children and animals. Keep/store only in original container.

Incompatible products

None known

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters

Chemical name	*British Columbia*	*Quebec*	*Ontario TWAEV*	*Alberta*
Petroleum distillates, hydrotreated light (64742-47-8)	TWA: 200 mg/m3 Skin	-	-	-

Appropriate engineering controls

Engineering measures

Apply technical measures to comply with the occupational exposure limits. Ensure adequate ventilation, especially in confined areas. When working in confined spaces (tanks, containers, etc.), ensure that there is a supply of air suitable for breathing and wear the recommended equipment.

Individual protection measures, such as personal protective equipment

Eye/Face Protection

If there is a potential for exposure to particles which could cause eye discomfort, wear chemical goggles.

Skin and Body Protection

Wear long-sleeved shirt, long pants, socks, and shoes.

Hand Protection

Protective gloves

Respiratory Protection

If exposure limits are exceeded or irritation is experienced, NIOSH/MSHA approved respiratory protection should be worn. Respiratory protection must be provided in accordance with current local regulations.

Hygiene measures

Clean water should be available for washing in case of eye or skin contamination. Wash skin prior to eating, drinking, chewing gum or using tobacco. Shower or bathe at the end of working. Remove and wash contaminated clothing before re-use. Launder work clothing separately from regular household laundry.

General information

If the product is used in mixtures, it is recommended that you contact the appropriate protective equipment suppliers.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance	Amber Liquid
Physical State	Liquid
Color	Amber
Odor	Faint hydrocarbon

Odor threshold pH Melting point/freezing point Boiling Point/Range Flash point Evaporation Rate Flammability (solid, gas) Flammability Limit in Air	No information available 3.75 Not applicable No information available 82.2 °C / 179.96 °F Tag Closed Cup No information available No information available No information available No information available No information available 8.61 lb/gal 1.033 @ 20°C Emulsifies No information available No information available No information available No information available No information available No information available No information available No information available No information available No information available No information available No information available No information available No information available
Upper flammability limit:	
Lower flammability limit:	
Vapor pressure	
Vapor density	
Relative density	
Specific gravity	
Water solubility	
Solubility in other solvents	
Partition coefficient	
Autoignition temperature	
Decomposition temperature	
Viscosity, kinematic	
Viscosity, dynamic	
Explosive properties	
Oxidizing properties	
Molecular weight	
Bulk density	

10. STABILITY AND REACTIVITY

Reactivity	None under normal use conditions
Chemical Stability	Stable under recommended storage conditions.
Possibility of Hazardous Reactions	None under normal processing.
Hazardous polymerization	Hazardous polymerization does not occur.
Conditions to avoid	Excessive heat, Extreme risk of explosion by shock, friction, fire or other sources of ignition
Incompatible materials	None known.
Hazardous Decomposition Products	Chlorine, Hydrogen chloride, Carbon oxides (COx), Aldehydes.

11. TOXICOLOGICAL INFORMATION

Product Information

LD50 Oral	998 mg/kg (rat)
LD50 Dermal	> 2000 mg/kg (rabbit)
LC50 Inhalation	> 4.3 mg/L 4 hr (rat)
Serious eye damage/eye irritation	Mildly irritating.
Skin corrosion/irritation	Moderately irritating.
Sensitization	Sensitizer

Chemical name	***LD50 Oral***	***LD50 Dermal***	***LC50 Inhalation***
Permethrin (52645-53-1)	3129 mg/kg (Rat) > 5000 mg/kg (Rat)	≥ 2000 mg/kg (Rabbit) > 2000 mg/kg (Rabbit)	> 2.09 mg/L (Rat) 4 h > 5.2 mg/L (Rat) 4 h
Petroleum distillates, hydrotreated light (64742-47-8)			

Information on toxicological effects

Symptoms Large toxic doses of the formulated product, administered by ingestion to animals, produced central nervous system effects with symptoms that include hypersensitivity to touch and sound, tremors, and clonic convulsions. Overexposure via inhalation produced symptoms such as squinting eyes, irregular and rattling breathing, and ataxia.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Chronic toxicity	Effects are expected to be similar to those that are seen with acute toxicity.
Mutagenicity	Permethrin: Not genotoxic in laboratory studies.
Carcinogenicity	Permethrin: Evidence of tumors in mice but is not considered relevant to humans.
Neurological effects	Permethrin: Clinical signs of neurotoxicity include altered motor activity and neurobehavioral effects, with no signs of histopathology.
Reproductive toxicity	Permethrin: No toxicity to reproduction in animal studies.
Developmental toxicity	Permethrin: Not teratogenic in animal studies.
STOT - single exposure	May cause damage to organs. See listed target organs below.
STOT - repeated exposure	May cause damage to organs through prolonged or repeated exposure. See listed target organs below.
Target organ effects	Permethrin: Central Nervous System (CNS).
Neurological effects	Permethrin: Clinical signs of neurotoxicity include altered motor activity and neurobehavioral effects, with no signs of histopathology.
Aspiration hazard	Potential for aspiration if swallowed.

Chemical name	***ACGIH***	***IARC***	***NTP***	***OSHA***
Permethrin 52645-53-1		Group 3		

Legend:
IARC (International Agency for Research on Cancer)
Group 3 - Not classifiable as to its carcinogenicity to humans

12. ECOLOGICAL INFORMATION

Ecotoxicity

Permethrin (52645-53-1)				
Active Ingredient(s)	Duration	Species	Value	Units
Permethrin	48 h EC50	Crustacea	0.001	mg/L
	72 h EC50	Algae	0.0125	mg/L
	21 d NOEC	Crustacea	0.039	µg/L
	96 hr NOEC	Algae	0.9	µg/L
	21 d NOEC	Fish	0.3	µg/L
	96 h LC50	Fish	5.3	mg/L
	96 h EC50	Algae	0.010	mg/L

Chemical name	***Toxicity to algae***	***Toxicity to fish***	***Toxicity to daphnia and other aquatic invertebrates***
Glyceral triacetate 102-76-1		48 h LC50: = 170 mg/L (Leuciscus idus) static 96 h LC50: = 2.2 mg/L (Lepomis macrochirus) static 96 h LC50: = 2.4 mg/L (Oncorhynchus mykiss) static 96 h LC50: = 45 mg/L (Pimephales promelas) flow-through 96 h LC50: 1120 - 1520 mg/L (Oncorhynchus mykiss) flow-through 96 h LC50: 1370 - 1670 mg/L (Pimephales promelas) flow-through 96 h LC50: 1480 - 1730 mg/L (Lepomis macrochirus) flow-through 96 h LC50: = 375 mg/L	48 h EC50: = 380 mg/L (Daphnia magna) 96 h LC50: = 4720 mg/L (Den-dronereides heteropoda)
Petroleum distillates, hydrotreated light 64742-47-8			
2-methylpropan-1-ol 78-83-1	48 h EC50: = 230 mg/L (Desmodesmus subspicatus)		48 h EC50: 1070 - 1933 mg/L (Daphnia magna) Static 48 h EC50: = 1300 mg/L (Daphnia magna)

		(Pimephales promelas) static	

Persistence and degradability Permethrin: Moderately persistent. Does not readily hydrolyze. Not readily biodegradable.

Bioaccumulation Permethrin: The substance has a potential for bioconcentration.

Mobility Permethrin: Immobile; Not expected to reach groundwater.

13. DISPOSAL CONSIDERATIONS

Waste disposal methods Pesticide wastes are toxic. Improper disposal of excess pesticide, spray mixture, or rinsate is prohibited. If these wastes cannot be disposed of by use according to label instructions, contact appropriate disposal authorities for guidance. Proper personal protective equipment, as described in Sections 7 and 8, must be worn while handling materials for waste disposal.

Contaminated Packaging Containers must be disposed of in accordance with local, state and federal regulations.
Refer to the product label for container disposal instructions.

14. TRANSPORT INFORMATION

DOT Not regulated for transportation if shipped in Non Bulk packaging. The classification below pertains to the shipment in Bulk packaging.

UN/ID no UN3082
Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
Hazard class 9
Packing Group III
Marine Pollutant Permethrin.
Description NA1993, Combustible liquid, n.o.s. (Aromatic hydrocarbons), III

TDG Classification below is only applicable when shipped by vessel and is not applicable when shipped by road or rail only.

UN/ID no UN3082
Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
Hazard class 9
Packing Group III
Marine Pollutant Permethrin.
Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (permethrin), 9, PGIII, Marine Pollutant

ICAO/IATA

UN/ID no UN3082
Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
Hazard class 9
Packing Group III
Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (permethrin), 9, PGIII, Marine Pollutant

IMDG/IMO

UN/ID no UN3082
Proper Shipping Name Environmentally hazardous substance, liquid, n.o.s.
Hazard class 9
Packing Group III
EmS No. F-A, S-F
Marine Pollutant Permethrin
Description UN3082, Environmentally hazardous substance, liquid, n.o.s. (permethrin), 9, PGIII, Marine Pollutant

15. REGULATORY INFORMATION

U.S. Federal Regulations

SARA 313 Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372:

Chemical name	*CAS-No*	*Weight %*	*SARA 313 - Threshold Values %*
Permethrin - 52645-53-1	52645-53-1	36.8	1.0

SARA 311/312 Hazard Categories

Acute health hazard	Yes
Chronic health hazard	Yes
Fire hazard	Yes
Sudden release of pressure hazard	No
Reactive Hazard	No

Clean Water Act

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42)

CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302):

Chemical name	*Hazardous Substances RQs*	*Extremely Hazardous Substances RQs*
2-methylpropan-1-ol 78-83-1	5000 lb 2270 kg	

FIFRA Information

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

CAUTION

Harmful if swallowed. Harmful if absorbed through the skin. Prolonged or frequent skin contact may cause allergic reactions in some individuals.
This product is extremely toxic to fish and aquatic invertebrates.

US State Regulations

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Chemical name	*New Jersey*	*Massachusetts*	*Pennsylvania*
Permethrin 52645-53-1	X	X	

International Inventories

Chemical name	*TSCA (United*	*DSL (Canada)*	*EINECS/ELINCS (Europe)*	*ENCS (Japan)*	*China (IECSC)*	*KECL (Korea)*	*PICCS (Philippines)*	*AICS (Australia)*

	States)							
Permethrin 52645-53-1			X	X	X	X	X	X
Petroleum distillates, hydrotreated light 64742-47-8	X	X	X		X	X	X	X
1,1'-Biphenyl, bis(1-methylethyl)- 69009-90-1	X	X	X	X	X			

Mexico - Grade Moderate risk, Grade 2

WHMIS Statement
This product has been classified in accordance with the Hazardous Products Regulations (HPR) and the SDS contains all the information required by the HPR.
WHMIS Hazard Class

D2B - Toxic materials



16. OTHER INFORMATION

NFPA	***Health Hazards*** 2	***Flammability*** 2	***Instability*** 0	***Special Hazards*** -
HMIS	***Health Hazards*** 2*	***Flammability*** 2	***Physical hazard*** 0	***Personal Protection*** X

****Indicates a chronic health hazard.***

NFPA/HMIS Ratings Legend Severe = 4; Serious = 3; Moderate = 2; Slight = 1; Minimal = 0

Revision date: 2019-06-03
Reason for revision: SDS sections updated
Disclaimer

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End of Safety Data Sheet